



UNIVERSAL BEAMS AND COLUMNS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	I _x		I _y		i _x		i _y		Z _x	Z _y		
in (mm)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W10 10 x 10 (254 x 254)	272	11,300	93.4	3,890	4.35	11.0	2.54	6.45	54.6	895	18.7	306
	303	12,600	103	4,290	4.37	11.1	2.56	6.50	60.0	983	20.6	338
	341	14,200	116	4,830	4.39	11.2	2.57	6.53	66.7	1,090	23.0	377
	394	16,400	134	5,580	4.44	11.3	2.58	6.55	75.7	1,240	26.4	433
	421	17,510	142	5,903	4.46	11.3	2.59	6.58	80.1	1,312	27.9	457
	455	18,900	154	6,410	4.49	11.4	2.60	6.60	85.9	1,410	30.1	493
	534	22,200	179	7,450	4.54	11.5	2.63	6.68	98.5	1,610	34.8	570
	542	22,580	181	7,518	4.55	11.6	2.63	6.68	99.7	1,634	35.2	577
	624	26,000	207	8,620	4.61	11.7	2.65	6.73	112.0	1,840	40.0	655
	716	29,800	236	9,820	4.66	11.8	2.68	6.81	126.0	2,060	45.3	742
W12 12 x 4 (305 x 102)	88.6	3,690	2.36	98.2	4.62	11.7	0.753	1.91	14.9	244	1.19	19.5
	103	4,290	2.82	117	4.67	11.9	0.773	1.96	17.1	280	1.41	23.1
	105.4	4,387	2.9	120	4.65	11.8	0.770	1.96	17.6	288	1.40	23.6
	130	5,410	3.76	156	4.82	12.2	0.822	2.09	21.3	349	1.88	30.8
	156	6,490	4.66	194	4.91	12.5	0.848	2.15	25.4	416	2.31	37.9
W12 12 x 5 (305 x 127)	172	7,162	8.1	337	4.83	12.3	1.05	2.67	28.8	472	3.30	54.6
	195.6	8,143	9.3	388	4.87	12.4	1.06	2.70	32.4	531	3.80	62.5
	228.3	9,504	11.1	460	4.92	12.5	1.08	2.75	37.4	612	4.50	73.5
W12 12 x 6 1/2 (305 x 165)	204	8,490	17.3	720	5.17	13.1	1.51	3.84	33.4	547	5.34	87.5
	204.1	8,496	16.6	691	5.06	12.8	1.44	3.66	34.1	559	5.10	83.6
	238	9,910	20.3	845	5.21	13.2	1.52	3.86	38.6	633	6.24	102
	238.4	9,924	19.8	824	5.11	12.9	1.47	3.73	39.4	646	6.10	100
	280.8	11,690	23.7	986.6	5.15	13.1	1.50	3.81	45.9	752	7.20	118
	285	11,900	24.5	1,020	5.25	13.3	1.54	3.91	45.6	747	7.47	122
W12 12 x 8 (305 x 203)	310	12,900	44.1	1,840	5.13	13.0	1.94	4.93	51.9	850	11.0	180
	350	14,600	50.0	2,080	5.15	13.1	1.94	4.93	58.1	952	12.4	203
	394	16,400	56.3	2,340	5.18	13.2	1.96	4.98	64.7	1,060	13.9	228
W12 12 x 10 (305 x 254)	425	17,700	95.8	3,990	5.23	13.3	2.48	6.30	70.6	1,160	19.2	315
	475	19,800	107	4,450	5.28	13.4	2.51	6.38	78.0	1,280	21.4	351
W12 12 x 12 (305 x 305)	533	22,200	174	7,240	5.28	13.4	3.02	7.67	87.9	1,440	29.1	477
	597	24,800	195	8,120	5.31	13.5	3.04	7.72	97.4	1,600	32.4	531
	662	27,600	216	8,990	5.34	13.6	3.05	7.75	107	1,750	35.8	587
	740	30,800	241	10,000	5.38	13.7	3.07	7.80	118	1,930	39.7	651
	789	32,840	256	10,670	5.40	13.7	3.08	7.82	125	2,084	42.2	691
	833	34,700	270	11,200	5.44	13.8	3.09	7.85	131	2,150	44.4	728
	933	38,800	301	12,500	5.47	13.9	3.11	7.90	145	2,380	49.3	808
	1,070	44,500	345	14,400	5.51	14.0	3.13	7.95	163	2,670	56.0	918
	1,221	50,840	390	16,230	5.59	14.2	3.16	8.03	183	2,991	63.1	1,034
	1,240	51,600	398	16,600	5.58	14.2	3.16	8.03	186	3,050	64.2	1,050
	1,430	59,500	454	18,900	5.66	14.4	3.19	8.10	209	3,420	72.8	1,190
	1,542	64,180	486	20,240	5.70	14.5	3.20	8.13	222	3,641	77.1	1,273
	1,650	68,700	517	21,500	5.74	14.6	3.22	8.18	235	3,850	82.3	1,350
	1,890	78,700	589	24,500	5.82	14.8	3.25	8.25	263	4,310	93.0	1,520
	2,140	89,100	664	27,600	5.89	15.0	3.28	8.33	292	4,780	104	1,700
	2,420	101,000	742	30,900	5.97	15.2	3.31	8.41	321	5,260	115	1,880
	2,720	113,000	828	34,500	6.06	15.4	3.34	8.48	353	5,780	127	2,080
	3,110	129,000	937	39,000	6.16	15.6	3.38	8.59	393	6,440	143	2,340
	3,550	148,000	1,050	43,700	6.29	16.0	3.42	8.69	435	7,130	159	2,610
	4,060	169,000	1,190	49,500	6.41	16.3	3.47	8.81	483	7,910	177	2,900
W14 14 x 5 (356 x 127)	199	8,280	7.00	291	5.54	14.1	1.04	2.64	29.0	475	2.80	45.9
	245	10,200	8.91	371	5.65	14.4	1.08	2.74	35.3	578	3.55	58.2